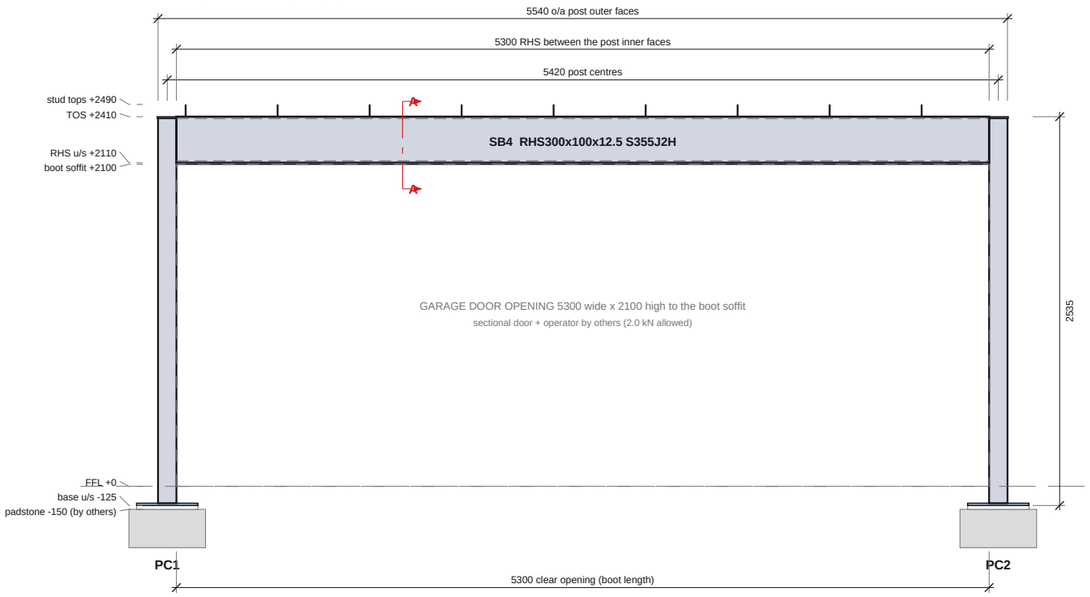
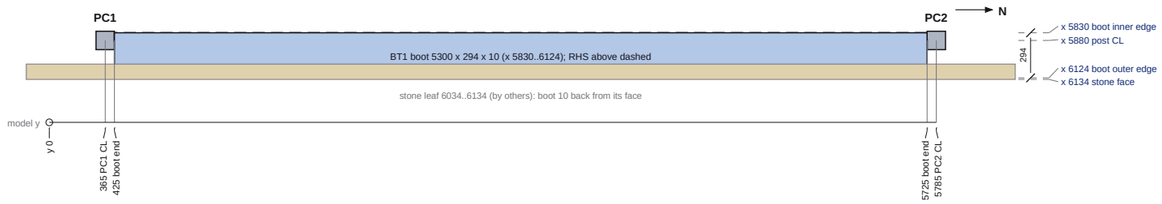
## 1 GP1 ELEVATION - LOOKING WEST FROM OUTSIDE

Scale 1:40 | levels above FFL; south (PC1) left, north (PC2) right



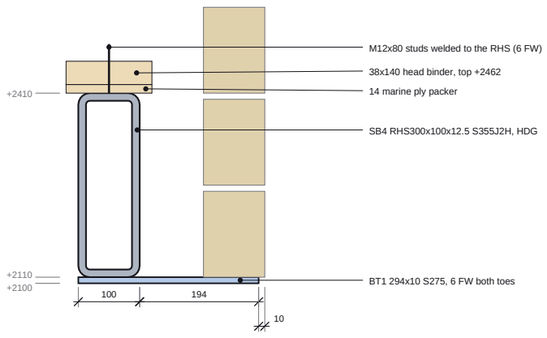

## 2 GP1 PLAN AT BOOT LEVEL

Scale 1:40 | outside (east) down; north right



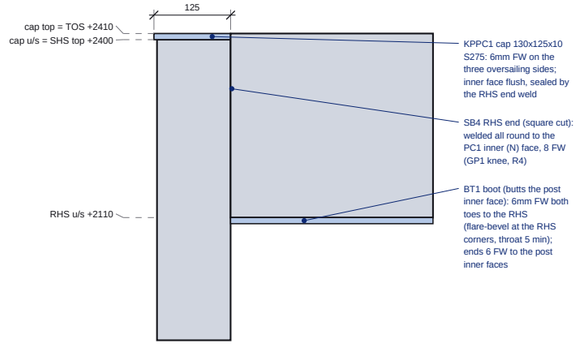

## 3 SECTION A-A

Scale 1:10 | looking north; inside left



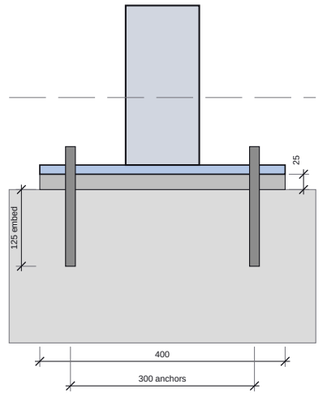

## 4 KNEE (PC1; PC2 HANDED)

Scale 1:10 | looking west from outside; north right




## 5 BASE (PC1 / PC2)

Scale 1:10 | elevation




BPPC1/BPPC2 400x200x15 S275 (200 across x 400 along the frame), 8mm FW all round
4 no. M16 8.8 HDG rod, R-KEM II, 125 embed, in Ø18 holes at x ±50 / y ±150 of the post CL (view along the frame: the two rows are in line)
25 HDG shims + non-shrink grout; padstone top -150 by others

### GP1 PARTS

| MARK | ITEM | LENGTH | KG |
|---|---|---|---|
| SB4 | RHS300x100x12.5 | 5300 | 383.2 |
| BT1 | PL 5300x294x10 S275 | | 122.22 |
| PC1 | SHS120x120x10 | 2510 | 84.6 |
| BPPC1 | PL 400x200x15 S275 | | 9.30 |
| KPPC1 | PL 130x125x10 S275 | | 1.28 |
| PC2 | SHS120x120x10 | 2510 | 84.6 |
| BPPC2 | PL 400x200x15 S275 | | 9.30 |
| KPPC2 | PL 130x125x10 S275 | | 1.28 |
| **GP1** | **welded assembly, HDG as one piece** | | **695.8** |

### NOTES

1. **APPROVED FOR CONSTRUCTION - the engineer approved the steelwork and the TLS details on 13.10.26; fabricate to these drawings.**
2. ALL STEELWORK SUBJECT TO SITE MEASURE BEFORE FABRICATION.
3. GP1 R4 - TLS design, approved by the engineer 13.10.26. Replaces S-0010's UB203x133x30: carries the 5300 door opening AND ~0.5 m of stone (main-house stone eaves).
4. LOADS (char.): trusses S-0012 DL 15 / IL 10 kN each end -> 5.4 / 3.6 kN/m; stone 0.5x0.1x23 = 1.15 kN/m; boot + timber + self weight 1.0 kN/m; door rails + operator 2.0 kN mid-span; wind 12 kN SLS / 18 kN ULS at the eaves.
5. RHS 300x100x12.5: I 9103 cm4, Wpl 806 cm3. ULS M 61 kNm vs Mpl 286 kNm. Deflection (SLS incl. door, pinned, L 5420) 6.9 mm = L/787 < 8 mm (client limit).
6. SHS 120x120x10 to TOS: N 44 kN + knee M 29.7 kNm vs Mpl 62 kNm; sway 6.8 mm = h/326.
7. KNEE (R4): RHS ends welded all round to the post inner faces, 8 FW: couple 29.7 / 0.288 = 103 kN per RHS wall vs 136 kN; post face (b1/b0 0.83) ~390 kN; torsion from the boot 0.84 kNm per knee.
8. BASE: 400x200x15, 54 clear of the stone leaf; N 44 kN, H 9 kN per base (ULS); pinned base assumed for sway. Boot toe ~0.22 kNm/m -> 13 MPa in the 10 plate.
9. By others: padstones on the garage return foundations (min 500 x 300), cavity tray / thermal bridge at the boot, 30 min fire protection (S-0011 9.02).

### REVISIONS

| REV | DATE | DESCRIPTION |
|---|---|---|
| C03 | 11.10.26 | Steel fabrication drawings - first issue |
| C04 | 13.10.26 | Approved by the engineer 13.10.26 - issued for construction |

TLS TIMBER FRAME DESIGN SUITE

**TLS TIMBER FRAME**
Design & Manufacture
Scan: 3D model + drawings

| CLIENT | PROJECT |
|---|---|
| MP Adams | MP Adams Plot 4, Eckington |

DRAWING TITLE

# GARAGE PORTAL GP1 ASSEMBLY

Welded goal-post frame SB4 + PC1 + PC2 with boot BT1 (TLS design, approved by the engineer)

| SCALE | SIZE | DATE | DRAWN |
|---|---|---|---|
| 1:40 / 1:10 | A3 | 13.10.26 | TS |

| STATUS | DRAWING No. | REV |
|---|---|---|
| CONSTRUCTION | P4-ST-GP1 | C04 |

Do not scale. All dimensions in mm. Check all dimensions on site.